Gauging Discrete Symmetry

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Abstract

This note is about gauging a discrete symmetry. We will first review what gauging is in a continuous case and then try to define a counterpart for a discrete symmetry. Then we try to derive some general consequences of theory after gauging.

Contents

Gauging is the case when we couple a QFT to a gauge field (often dynamical, but some people may also call background gauge field as gauging). Apart from its importance of studying interaction between QFTs, gauging is also a powerful tool to study the symmetry of a QFT.

1 How to Gauge a Continuous Symmetry?

First we review some basics of coupling a QFT to a gauge field. Of course, we can just do the minimal coupling and be happy with it. However, to make it more general, we should not assume we always have a conserved current or even a classical lagrangian description. Then, how can we do it?

1.1 Symmetry Characterized by Generators

When we want to pin down a symmetry in QFT, the usual tools are the conserved current and the conserved charge. But these come from Noether's theorem, which already assumes two things: the symmetry is continuous, and it's invertible (so it forms a group). Neither is really essential to being a symmetry — so current and charge can't capture everything.

The Ward identity does better. A symmetry shows up as non-trivial constraints on correlation functions, and that's exactly what the Ward identity encodes. No group structure needed, so non-invertible symmetries are fine too. The catch: a Ward identity is a constraint, not an object you can build and play with. Moreover, having a constraint on correlation function doesn't necessarily mean we have a symmetry, for example, if a symmetry has an anomaly, the correlation function indeed satisfy some constraints, but we think of it as the symmetry is broken.

So what we'd really like is an "object-like" thing — like a current or charge — that's general enough and accurate enough to describe any symmetry. Good news: it exists. It's the **symmetry generator**, and the trick is that it's a topological operator. That's the property that sets us free from continuity and invertibility, and it's what we'll build the definition around.

In what follows, we'll let our intuition about continuous symmetries lead the way — and watch the conserved charge turn into a topological operator.

1.1.1 Classical Definition

We first take an example of a classical field theory with a U(1) symmetry (say a complex scalar field). Noether's theorem tells us that there is a conserved current j satisfying the conservation equation:

$$d * j = 0 \tag{1}$$

And we can define a conserved charge by integrating the current over a spatial slice:

$$Q = \oint_{\Sigma} * j \tag{2}$$

where Σ is a spatial slice. Now we can try to integrate the current over a codimension-1 manifold M_{d-1}, which is not necessarily a spatial slice, and define a charge as:

$$Q(M_{d-1}) = \oint_{M_{d-1}} * j \tag{3}$$

and a related classical quantity we called the **symmetry generator**:

$$U(M_{d-1}) = \exp(i\alpha Q(M_{d-1})) \tag{4}$$

One can prove that these objects are topological, which means that they are invariant under continuous deformation of the manifold M_{d-1}, as long as we don't cross any strange stuffs. This can be proven with the stokes theorem:

$$Q(M_{d-1}) = \oint_{M_{d-1}} *j = \oint_{M'_{d-1}} *j + \oint_{B_d} d*j = Q(M'_{d-1}) \tag{5}$$

where B_d is the manifold connecting M_{d-1} and M'_{d-1} satisfying:

$$\partial B_d = M_{d-1} \sqcup M'_{d-1} \tag{6}$$

1.1.2 Quantum Symmetry Generator

Now we generalize the classical object we define above to a quantum one. Notice that the world is NOT classical, it is definitely true that we can have some quantum object without classical counterparts. Namely can't be derived from quantizing some classical object using single quantization scheme (eg. canonical quantization), and the symmetry generator is such an object.

Thus, we don't derive a quantum theory from a classical one, but use different quantization scheme and results to justify how a quantum symmetry generators exit what properties should they have.

The properties of **quantum symmetry generators** (and justifications) are:

1. They are **Topological Operators**. This can be justified by the classical property of independent of continuous deformation, plugging them into a path integral won't change this fact. Moreover, as a quantum theory, in the canonical picture if we want to define these symmetry generators as line operators, they have to commute with the Energy-Momentum tensor, which generates the spacetime translation.
2. If inserted in a time slice, they act as an **operators on the Hilbert space** in the canonical formalism. This can be justified by analogizing the ordinary symmetry charge defined on a time slice.
3. If inserted in vertically in the path integral, they behave as a **defect, which twisted the Hilbert Space**. This can't be justified straightforwardly, but a few lines of calculation in the path integral formalism, for example, consider a free massless complex scalar field theory in 2D Euclidean spacetime:

Insert the defect along S^1_τ at $x = 0$ into the Euclidean path integral:

$$Z_\theta = \int \mathcal{D}\Phi\, \mathcal{D}\Phi^\dagger\, e^{-S[\Phi]} \underbrace{\exp\left(i\theta \int_0^\beta d\tau\, j_x(\tau, 0) \right)}_{U_\theta}, \quad j_x = i\big(\partial_x \Phi^\dagger\big)\Phi - i\Phi^\dagger \partial_x \Phi. \tag{7}$$

The insertion couples the current to a source localized at $x = 0$. Writing it as $\exp\big(i \int d^2x\, A_\mu j^\mu\big)$ requires

$$A_x(\tau, x) = \theta\, \delta(x), \quad A_\tau = 0. \tag{8}$$

This promotes $\partial_\mu \Phi \to \big(\partial_\mu - iA_\mu\big)\Phi$ in the action. Compute the field strength:

$$F_{\tau x} = \partial_\tau A_x - \partial_x A_\tau = \partial_\tau(\theta\, \delta(x)) = 0. \tag{9}$$

A flat connection carries no local physics; it can be removed by a gauge transformation $\Phi \to e^{i\alpha(x)}\Phi$, $A_x \to A_x + \partial_x \alpha$. Choose

$$\alpha(x) = -\theta \int_0^x \delta(x')\, dx' = -\theta\, \Theta(x), \tag{10}$$

so that $A_x + \partial_x \alpha = \theta\delta(x) - \theta\delta(x) = 0$. Define $\tilde{\Phi} = e^{i\alpha(x)}\Phi = e^{-i\theta\Theta(x)}\Phi$. Then $\tilde{\Phi}$ obeys the **standard free action** $S[\tilde{\Phi}]$ with no defect, and the measure is invariant ($|e^{i\alpha}| = 1$, Jacobian $=$ 1). The original periodic field $\Phi(\tau, x + 2\pi) = \Phi(\tau, x)$ forces, on the redefined field, a monodromy across one spatial period:

$$\tilde{\Phi}(\tau, x + 2\pi) = e^{-i\theta\Theta(x+2\pi)}\Phi(\tau, x + 2\pi) = e^{-i\theta(\Theta(x)+1)}\Phi(\tau, x) = e^{-i\theta}\,\tilde{\Phi}(\tau, x). \tag{11}$$

Equivalently, the field carrying the defect satisfies the **twisted** boundary condition

$$\Phi_\theta(\tau, x + 2\pi) = e^{i\theta}\,\Phi_\theta(\tau, x). \tag{12}$$

The defect insertion is traded for a boundary twist:

$$Z_\theta = \int_{\Phi(x+2\pi)=e^{i\theta}\Phi(x)} \mathcal{D}\Phi\, e^{-S[\Phi]} = \mathrm{tr}_{\mathcal{H}_\theta(S^1_x)}\, e^{-\beta H}. \tag{13}$$

Inserting U_θ along Euclidean time $=$ tracing over the Hilbert space quantized with the twisted boundary condition.

> **Note**
>
> 2 and 3 properties are called the operator/defect principles [1]

4. When going through local operators, they act as a **symmetry transformation on the local operators**. This can be justified by the Ward identity.

For example, consider inserting the symmetry generator on a manifold that wrapped around the local operator, for a U(1) 0-form symmetry and a local operator. We define wrapping around the local operator in the canonical picture as:

$$U_\alpha(M_{d-1})\mathcal{O}_q(x)U_{-\alpha}(M'_{d-1}) \tag{14}$$

We can calculate the correlation function of this object using the Ward Identity.

$$U_\alpha(M_{d-1})\mathcal{O}_{q(x)}U_{-\alpha}(M'_{d-1}) = e^{i\alpha\int_{N_d} d^d y\, \partial^\mu j_\mu(y)}\mathcal{O}_{q(x)}. \tag{15}$$

Remember, the Ward-Takahashi identity:

$$\partial^\mu j_{\mu(y)}\mathcal{O}_{q(x)} = q\,\delta^{(d)}(x - y)\mathcal{O}_{q(x)} \tag{16}$$

By expanding the exponential and using the Ward-Takahashi identity, we have:

$$= \sum_{n=0}^{\infty} \frac{(i\alpha)^n}{n!}\left(\int_{N_d} d^d y\, \partial^\mu j_{\mu(y)}\right)^n \mathcal{O}_{q(x)} = \sum_{n=0}^{\infty} \frac{(i\alpha q)^n}{n!}\left(\int_{N_d} d^d y\, \delta^{(d)}(x - y)\right)^n \mathcal{O}_{q(x)}. \tag{17}$$

Since $x \in N_d$, the integral $= 1$, so the sum resums to

$$= \sum_{n=0}^{\infty} \frac{(i\alpha q)^n}{n!}\mathcal{O}_{q(x)} = e^{iq\alpha}\mathcal{O}_{q(x)}. \tag{18}$$

1.1.3 A Slogan for Symmetry

The current/charge picture is tied to continuous symmetries with a classical Lagrangian. To go beyond it, we adopt the following slogan.

> **Definition 1.1** **Symmetry Slogan**
>
> A symmetry is characterized by the existence of a topological operator — a *symmetry generator.*

This is a powerful and sensible generalization. As we will see, it applies equally to continuous symmetries, discrete symmetries, and even symmetries without an inverse, the so-called *non-invertible symmetries.*

1.2 (Higher Form) Gauge Field

Now we have a grasp of what is a symmetry, we turn to the other question of gauging: what is a gauge field?

1.2.1 What is a Gauge Field?

Naively, we define a gauge field as a Lie algebra valued one form field:

$$A = A_\mu dx^\mu \text{ where } A_\mu \in \mathfrak{g} \tag{19}$$

However, this is generally not the case.

- **A gauge field may not be globally defined.**

A classical and experimental example is the AB effect of U(1) gauge field, where we know that:

$$\exp\left(i \oint_C A \,\mathrm{d}l\right) \tag{20}$$

and is exactly an experimental observable. Thus we need to rethink the definition of gauge field. The critical point is that a gauge field may not be defined globally, but only on patches, while field on different patches are related by a suitable gauge transformation.

Moreover, from demanding consistency in AB phase, and a globally well defined curvature (indeed we measure electric and magnetic fields and don't see any singularity) naively we can have:

$$\oint_S F = 0 \tag{21}$$

But in fact this is too strong, we in fact only need to have a quantization condition on the flux of the gauge field:

$$\oint_S F \in 2\pi\mathbb{Z} \tag{22}$$

To have the theory well defined and the AB phase (or called Wilson loop) as an observable. However, in the case that the flux is non-trivial, an important phenomenon is that the gauge field is not globally defined, but only on patches, and the transition function between patches is non-trivial.

> **Note**
>
> In fact the non-trivial flux is physically interpreted as the integrating around the surface with magnetic charge inside. Experimentally, we haven't observed magnetic charge in the universe, but it is good to leave the possibility of their existence, which makes our life more interesting.

- **A Gauge Transformation Parameter is not Globally Defined**

Moreover, gauge field admits gauge transformation:

$$A \to A + d\Lambda \tag{23}$$

that doesn't change the physics. However, the gauge transformation parameter Λ may not be globally defined, but only on patches. The experiments and some deduction tells us that one can never tell whether a solenoid has a magnetic flux Φ or $\Phi + \Phi_0$, where Φ_0 is the flux quantum. Thus, we should view that the transformation of adding a flux quantum as a gauge transformation. However, we may notice that such gauge transformation doesn't have a globally well defined gauge transformation parameter, but only on patches:

$$\Lambda = \frac{\Phi_0}{2\pi}\phi, \quad \phi \in [0, 2\pi), \tag{24}$$

where ϕ is the azimuthal angle around the solenoid. Indeed,

$$d\Lambda = \frac{\Phi_0}{2\pi}d\phi \tag{25}$$

is a globally well-defined closed 1-form, yet Λ itself jumps by Φ_0 upon encircling the solenoid:

$$\Lambda(\phi = 2\pi) - \Lambda(\phi = 0) = \Phi_0, \tag{26}$$

so it is single-valued only on patches, not globally. The holonomy picks up exactly one flux quantum,

$$\oint d\Lambda = \Phi_0, \tag{27}$$

which shifts the Aharonov–Bohm phase by $q\Phi_0 = 2\pi$ (for unit charge), leaving all physical observables invariant. One should notice that though Λ is note globally defined, $d\Lambda$ is globally defined, and the flux of $d\Lambda$ is quantized.

1.2.2 Mathematical Characterization

A mathematical framework, namingly **fibre bundle** is developed to characterize all these features of gauge field (which I wish to introduce in another note). Now to make things simple, we try to construct a gauge field step by steps:

- **Prepare Patches of Principle G Bundle**:

1. Given a manifold M_d choose a set of open over $\{U_i\}$
2. On each patch U_i we "choose a trivialization of the Pinciple G-bundle". This is kinds of equivalent to another fancy name, which is choosing a **section** of the bundle, which is a map from the base space to the total space of the bundle:

$$s_i : U_i \to P \tag{28}$$

well we can just understand as mapping each point in the patch to a group element and its only defined locally. Physically this is equivalent to choosing a gauge, as discussed before, we can't choose a global gauge, thus we can only choose a local gauge on each patch.

3. On the overlap of two patches U_i and U_j, we have a transition function:

$$g_{ij} : U_i \cap U_j \to G \tag{29}$$

this relates the two trivialization on the two patches, and we have:

$$s_i = g_{ij}s_j \tag{30}$$

4. On a triple overlap of three patches U_i, U_j and U_k, we have a consistency condition:

$$g_{ij}g_{jk}g_{ki} = 1 \tag{31}$$

- **Construct Gauge Field**:

gauge field can be understood as a connection on the principle G-bundle after a pullback by the section we choose. But we can just understand it as a well defined Lie algebra valued one form on each patch:

$$A^{(i)} \in \Omega^1(U_i) \otimes \mathfrak{g} \tag{32}$$

and on the overlap of two patches, we have a gauge transformation:

$$A^{(j)} = g_{ij}^{-1} A^{(i)} g_{ij} + g_{ij}^{-1} dg_{ij} \tag{33}$$

Two important objects of the principle bundle are the curvature and the holonomy of the gauge field, which are defined as:

$$F^{(i)} = dA^{(i)} + A^{(i)} \wedge A^{(i)} \tag{34}$$

$$\mathrm{Hol}(C) = \exp\left(i \oint_C A\right) \tag{35}$$

physically, they correspond to the field strength and the AB phase of U(1) gauge field.

- **Quantization of Flux**:

With suitable consistent mathematical construction of gauge field from fibre bundles, we can see that quantized flux arises naturally from mathematical properties of fibre bundles.

- Quantization of U(1) flux is related to the first Chern class of the U(1) bundle, which is an integer.

$$c_1 = \frac{1}{2\pi} \oint_S F \in \mathbb{Z} \tag{36}$$

- Quantization of SU(N) non-abelian flux is related to the second Chern class of the non-abelian bundle, which is also an integer.

$$c_2 = \frac{1}{8\pi^2} \oint_S \mathrm{Tr}(F \wedge F) \in \mathbb{Z} \tag{37}$$

1.2.3 Higher Form U(1) Gauge Field

This note we try to touch some generalization of symmetry to higher form. Thus, we need to define what is a higher form gauge field. A theorem shows that:

> **Theorem 1.2** **Abelian Higher Form Symmetry**
>
> A higher form symmetry is always an abelian symmetry, thus the gauge field of a higher form symmetry is always an abelian gauge field.

To make life simple, we here just consider **U(1) higher form symmetry** and **U(1) higher form gauge field**, which is a common case.

For a p-form symmetry, the current is a (p+1)-form j, thus it should couple to a (p+1)-form U(1) gauge field A as:

$$S = \int_{M_d} A \wedge * j \tag{38}$$

where A is a (p+1)-form gauge field, and the gauge transformation (no matter large or small) is:

$$A \to A + d\Lambda_p \tag{39}$$

And here Λ_p is a **p-form Gauge Field** itself!!! For A as a (p+1)-form gauge field, the field strength is a (p+2)-form, that are quantized under the following condition:

$$F = dA, \quad \oint_{S_{p+2}} F \in 2\pi\mathbb{Z} \tag{40}$$

it is quantized for any p+2 dimensional closed manifold S_{p+2}. Moreover, we should notice that the gauge transformation is also a gauge field. Thus there is a "gauge transformation of gauge transformation", which is a (p-1)-form gauge field Λ_{p-1}:

$$\Lambda_p \to \Lambda_p + d\Lambda_{p-1} \tag{41}$$

and the gauge transformation have a quantized flux condition as well:

$$\oint_{S_{p+1}} d\Lambda_p \in 2\pi\mathbb{Z} \tag{42}$$

Notice that just like the 1-form case Λ_p may not be globally defined, but $d\Lambda_p$ is globally defined.

1.3 How to Couple to a Gauge Field?

Now we start with the true topic, Gauging. By gauging a symmetry, we mean coupling the theory to a gauge field and make it dynamical. This section is to point out an observation. Namely, coupling to a gauge field is equivalent to inserting the symmetry generator properly.

In the section Section 1.1.2, we have notice this fact, inserting the symmetry generator along time direction gives us the coupling to a background gauge field. This should be a point to help us generalize gauging to discrete symmetries, since we don't really have a gauge field or a current, but we have a symmetry generator in discrete case.

Now we first understand this fact in the continuous case. Say consider a U(1) 0-form symmetry, we can insert the symmetry generator as:

$$U_\alpha(M_{d-1}) = \exp\left(i\alpha \int_{M_{d-1}} *j\right) \tag{43}$$

We can see that this object is exactly the same as coupling to a background gauge field A with:

$$A = \alpha\omega_{M_{d-1}} \tag{44}$$

where $\omega_{M_{d-1}}$ is a delta function one form localized on the manifold M_{d-1}, which is the Poincare dual of M_{d-1}. We can rewrite the symmetry generator as:

$$U_\alpha(M_{d-1}) = \exp\left(i \int_{M_d} A \wedge *j\right) \tag{45}$$

Thus, we can see a general fact that:

- Coupling to a background gauge field is equivalent to inserting the symmetry generator in a proper way.

> **Remark**
>
> The calculation here is classical, but we can understand it as quantum by plugging it into the path integral.

Also for gauge invariant operators in background gauge field, we can understand its correlation function as a theory without background gauge field but with some symmetry generator inserted.

For example, consider a gauge invariant local operator in U(1) symmetry theory coupling to a background gauge field A. If the local operator transform non-trivially under the U(1) symmetry, say:

$$\mathcal{O}_q \to e^{iq\alpha}\mathcal{O}_q \tag{46}$$

Then it is not a gauge invariant operator, consider a 2 point function correlation function:

$$\langle \mathcal{O}_q(x)\mathcal{O}_{-q}(y)\rangle \tag{47}$$

it is no longer gauge invariant under a gauge transformation $A \to A + d\alpha$, since the local operator transform non-trivially. However, we can insert a Wilson line connecting the two local operators, which is defined as:

$$W_q(C) = \exp\left(iq\oint_C A\right) \tag{48}$$

Then the correlation function:

$$\langle \mathcal{O}_q(x)W_q(C)\mathcal{O}_{-q}(y)\rangle \tag{49}$$

is gauge invariant. Then, if we consider one operator move around, the Wilson line should also move with it, thus the gauge invariant correlation function become:

$$\mathcal{O}_q(x) \to \exp\left(iq\int_\gamma A\right)\mathcal{O}_q(x) \tag{50}$$

In the case of gauge field of:

$$A = \alpha\omega_{M_{d-1}} \tag{51}$$

we can see that the gauge invariant correlation function behaves as:

$$\mathcal{O}_q(x) \to \exp(iq\alpha)\mathcal{O}_q(x) \tag{52}$$

which is exactly the bahavior of the local operator going through a symmetry generator.

1.4 Anomaly

For a system with a lagrangian description, we may expect a symmetry of the lagrangian may not be a symmetry of the QFT, say, the corresponding Ward identity is not satisfied. This is what we call an **anomaly**.

1.4.1 Definition and Classification

Due to the discussion of symmetry and background gauge field, we can give a definition from the perspective of gauging.

Definition 1.3 **Anomaly**

We say a p-form symmetry has an anomaly if we couple the theory to a background gauge field (which is of course a (p+1)-form gauge field):

$$Z[A_{p+1}] = \int \mathcal{D}\phi^a e^{iS[\phi^a, A_{p+1}]}. \tag{53}$$

and find that the theory is not gauge invariant:

$$Z[A_{p+1} + d\lambda_p] = e^{-i\int \mathcal{A}[A_{p+1}, \lambda_p]} \times Z[A_{p+1}]. \tag{54}$$

There are some classification and nouns for anomalies.

- **'t Hooft Anomaly**: the anomaly vanishes when the gauge field we couple to is turned off.
- **Gauge Anomaly**: no matter the gauge field is on or off, the symmetry is always broken.

There is also something called a **mixed 't Hooft anomaly**, which is when the theory have 2 symmetries, and if we couple one of them to a background gauge field, then if we want to couple the other one to a background gauge field, it won't be gauge invariant.

There are two examples of anomalies:

- In QED, there is a gauge anomaly, know as the ABJ anomaly.

$$\mathcal{L} = \frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} + i\overline{\Psi}(\partial_\mu - ia_\mu)\gamma^\mu \Psi. \tag{55}$$

The axial U(1) symmetry is broken by the ABJ anomaly, Ward identity is not satisfied, and becomes:

$$d * j^A = \frac{1}{4\pi^2} f \wedge f \tag{56}$$

- In pure 4D Maxwell theory, there is a mixed 't Hooft anomaly between the electric and magnetic 1-form symmetries:

$$j^E = \frac{1}{e^2} f, \qquad j^M = \frac{1}{2\pi} * f. \tag{57}$$

1.4.2 Anomaly Inflow

One trick to control the anomaly is to introduce a theory with an extra dimension, and have the original theory living on the boundary of the extra dimension manifold.

Consider a theory with anomaly that coupled to a background gauge field, living on the manifold $\mathcal{M}_d$:

$$Z[A_{p+1}] = \int \mathcal{D}\phi^a e^{iS[\phi^a, A_{p+1}]} \tag{58}$$

We can view the anomalous partition function not as a number, but as section of a complex line bundle or a vector in an one dimensional Hilbert space without a canonical choice of basis:

$$Z[A_{p+1}] \in \mathcal{H}_{\mathcal{M}_d} \tag{59}$$

Then we can think of this one dimensional Hilbert space as the Hilbert space of some d+1 dimensional QFT with the boundary $\mathcal{M}_d$. We can use the d+1 dimensional model to characterize the anomaly of the original theory, and this is what we call **anomaly inflow**.

- **Example Construction**

There is one explicit construction of the d+1 dimensional theory. We introduce a theory living on a manifold $\mathcal{N}_{d+1}$ with the boundary:

$$\mathcal{M}_d = \partial \mathcal{N}_{d+1} \tag{60}$$

The theory should have the following properties:

- The Hilbert space of the theory on $\mathcal{M}_{d+1}$ is one dimensional.
- The action of the theory, say: $\hat{A}\left[A_{p+1}\right]$ should satisfy:

$$d\mathcal{A}\left[A_{p+1}, \lambda_p\right] = \delta_{\lambda_p} \hat{A}\left[A_{p+1}\right], \tag{61}$$

Then if we consider a theory of:

$$\hat{Z}\left[A_{p+1}\right] = Z\left[A_{p+1}\right] \times e^{i \int_{N_{d+1}} \hat{A}\left[A_{p+1}\right]}, \tag{62}$$

Then it is invariant under gauge transformation of the background gauge field and we see that the anomaly is "controlled" by the d+1 dimensional theory.

2 How to Gauge a Discrete Symmetry?

Now we turn to the main topic of this note, gauging a discrete symmetry. To do this we may first define what we mean by a discrete gauge field, and a discrete symmetry of a QFT.

2.1 Discrete Symmetry

Wigner Theorem tells us that a Quantum theory has a symmetry is given by a Unitary or Anti-Unitary operator, forming a representation of the symmetry group on the Hilbert space. We can characterize the symmetry with these operators.

A discrete symmetry is a symmetry whose symmetry group is a discrete group, which can be finite or infinite. The Hilbert space forms a Unitary or Anti-Unitary representation of the symmetry group, and those operator

- **Group Structure**:
$$U(g_1)U(g_2) = U(g_1 g_2) \tag{63}$$
- **Conserved**:
$$U(g)HU(g)^{-1} = H \tag{64}$$
- **Generate transformation**:
$$U(g)O(x)U(g)^{-1} = R(g)O(x) \tag{65}$$
- **Ward Identity**: they also have some Ward identities, which can be used to constrain the correlation function. A simple example is the $\mathbb{Z}_2$ symmetry of free scalar field theory, which acts as $\varphi(x) \to -\varphi(x)$, and the Ward identity tells us that the correlation function of odd number of φ must vanish.
- ...

Remember the slogan we introduced before, we can generalize to characterize symmetries with symmetry generators. Imposing consistency conditions and observing the relation with Ward identities, we can find out all these symmetry generators and identify them with symmetries.

As far as I know, in 2D QFT these generalized symmetries can be described by Topological Defect Lines systematically [2], satisfying a series of characteristic properties. It turns out that these symmetries doesn't have a group structure, but rather a structure of fusion category. For higher dimensional QFT, the story is more complicated and general.

2.2 Discrete Gauge Field

Another problem is what is a discrete gauge field and what mathematical language characterize these objects. For discrete symmetry we don't have a Lie algebra, the Lie algebra of a discrete group is 0 thus:

$$A = 0, \quad F = 0 \tag{66}$$

Naively we only have trivial objects that are meaningless. However, we remember that a gauge field may not be globally defined, but only on patches. Though we have 0 for the gauge field locally, we may have non-trivial holonomy when going through patches:

$$\mathrm{Hol}(C) = \exp\left(i \oint_C A\right) = g \in G \tag{67}$$

In fact to characterize this data, we don't need to introduce the gauge field but the transition function between patches:

$$g_{ij} : U_i \cap U_j \to G \tag{68}$$

2.2.1 $\mathbb{Z}_n$ Gauge Field from BF Theory

One of the simplest examples of a discrete gauge field is the $\mathbb{Z}_n$ gauge field. A construction of this is that they can be obtained from a BF theory with higher form U(1) gauge field. Consider the following action:

$$S_{\mathrm{BF}} = \frac{n}{2\pi} \int_{M_d} a_{p+1} \wedge db_{d-p-2} \tag{69}$$

where a and b are U(1) higher form gauge fields. We finally can see that if b field is dynamical, then the a field is a $\mathbb{Z}_n$ gauge field, and if a field is dynamical, then the b field is a $\mathbb{Z}_n$ gauge field. We here take b as example, and the other is just the same under a integral by part.

- **Gauge Invariant of BF Theory**

We first need to justify that the BF theory is a valid gauge theory that has gauge invariance. We may notice that the action is not gauge invariant but the partition function is. Consider the gauge transformation:

$$a_{p+1} \to a_{p+1} + d\lambda_p, \qquad b_{d-p-2} \to b_{d-p-2} + d\hat{\lambda}_{d-p-3}. \tag{70}$$

We take the gauge transformation of a field then we have:

$$\delta_\lambda S_{BF} = \frac{n}{2\pi} \int_{M_d} d\lambda_p \wedge db_{d-p-2}, \tag{71}$$

Naively, this is 0, due to the stokes theorem, we can rewrite it as a total derivative. However, we should notice that the stokes theorem may not be valid here due to we are considering gauge field but not a globally well defined one form.

Thus, here $d\lambda_p$ and db_{d-p-2} are some forms that of course closed but **not necessarily Exact**. Don't be fooled by the fact they can write as a total derivative, they are not total derivatives of a globally well defined form, but a gauge field. For such integral of closed but not exact forms, there is also a way to control the integral of wedge product of them.

Mathematically, due to the exact of the field strength and quantization of flux, we can write:

$$\int_{M_d} d\lambda_p \wedge db_{d-p-2} = (2\pi)^2 \mathbb{Z} \tag{72}$$

as a result.[1] Thus, we see that the action is not gauge invariant, but we can make the partition function gauge invariant by demanding:

$$n \in \mathbb{Z} \tag{73}$$

[1] In the lecture note [3], the argument of this is confusing and some following discussions are wrong. Please see the appendix for Justification of this Section 4.1

3 Consequence Under Gauging

4 Appendix

4.1 A. BF Theory Gauge Invariance

In the discussion of BF theory gauge in Section 2.2.1, we have a critical step that the integral of the wedge product of two closed but not exact forms is quantized. Here we give a justification of this fact. Consider the integral of two gauge fields a and b:

$$\int_{M_d} da_p \wedge db_{d-p-2} \tag{74}$$

One critical observation is that this integral depends only on the de Rham cohomology class of da and db. This is because, if we shift da by an exact form, say $da \to da + d\lambda$, then the integral shift by:

$$\int_{M_d} d\lambda \wedge db_{d-p-2} = \int_{M_d} d\big(\lambda \wedge db_{d-p-2}\big) = 0 \tag{75}$$

The second step is valid for stokes theorem, since $\lambda \wedge db$ is a globally well defined form.

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